Microsoft Elevate

CAPSTONE PROJECT

Dark Store AI: Hyper-Local Demand Forecasting

Presented By

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Problem Statement:

Currently, quick commerce platforms like Blinkit or Zepto are introduced in many urban areas for the rapid fulfillment of 10-minute delivery promises. It is important to make the right inventory available and accessible to the public at the right time, as it lessens financial waste from overstocking and prevents customer frustration from stockouts. Eventually, providing these localized "dark stores" with a stable supply of goods becomes a major concern. The crucial part is the prediction of the exact inventory count required at each specific hour for the stable supply of this highly localized chain.

Proposed Solution:

The proposed system aims to address the challenge of predicting the required inventory count at each hour to ensure a stable supply of goods in dark stores. This involves leveraging data analytics and machine learning techniques to forecast demand patterns accurately. The solution will consist of the following components:

Data Collection:

- Gather historical data on customer orders, including time, date, product categories, and other relevant factors.
- Utilize real-time data sources, such as weather conditions, local events, and holidays, to enhance prediction accuracy.

Data Preprocessing:

- Clean and preprocess the collected data to handle missing values, outliers, and inconsistencies.
- Feature engineering to extract relevant features from the data that might impact localized delivery demand.

Deployment:

- Develop a user-friendly interactive dashboard that provides real-time predictions for inventory counts at different hours.
- Deploy the solution on a scalable and reliable platform, considering factors like data integration, response time, and manager accessibility.

System Approach:

Phase 1: Data Processing & Machine Learning

- Environment: Google Colab
- Libraries: Python, Pandas, Scikit-Learn
- Goal: Clean raw historical data, engineer time-based features, and train the predictive model.

Phase 2: Business Intelligence & Deployment

- Environment: Microsoft Power BI
- Tools: Power Query, DAX (Data Analysis Expressions)
- Goal: Transform the machine learning output into an interactive, visually appealing interface for decision-makers.

Google Colab Link

Algorithm & Deployment:

- **Algorithm Selection: XGBoost Regressor.** This advanced machine learning model was chosen over traditional time-series models (like ARIMA) because it excels at finding complex, non-linear patterns in tabular data—making it perfect for predicting sudden, volatile rush-hour spikes in dark stores.
- **Data Input:** The model relies on structured, time-based features extracted from the raw data. The specific inputs include: historical order volumes, the specific hour of the day (0-23), the day of the week, and a binary weekend/weekday indicator.
- **Training Process:** The algorithm is trained using the past 30 days of historical order data. To ensure accuracy, we utilized a chronological train-test split (rather than random splitting) so the model strictly learns forward-facing time trends without accidentally "peeking" at future data.
- **Prediction Process & Deployment:** During the prediction phase, the trained model evaluates the calendar parameters for the upcoming 7 days and outputs the precise inventory count needed for each future hour.

These predictions are then deployed into an interactive Power BI dashboard, utilizing native AI tools to automatically detect traffic anomalies and explain the drivers behind sudden demand spikes.

Result:

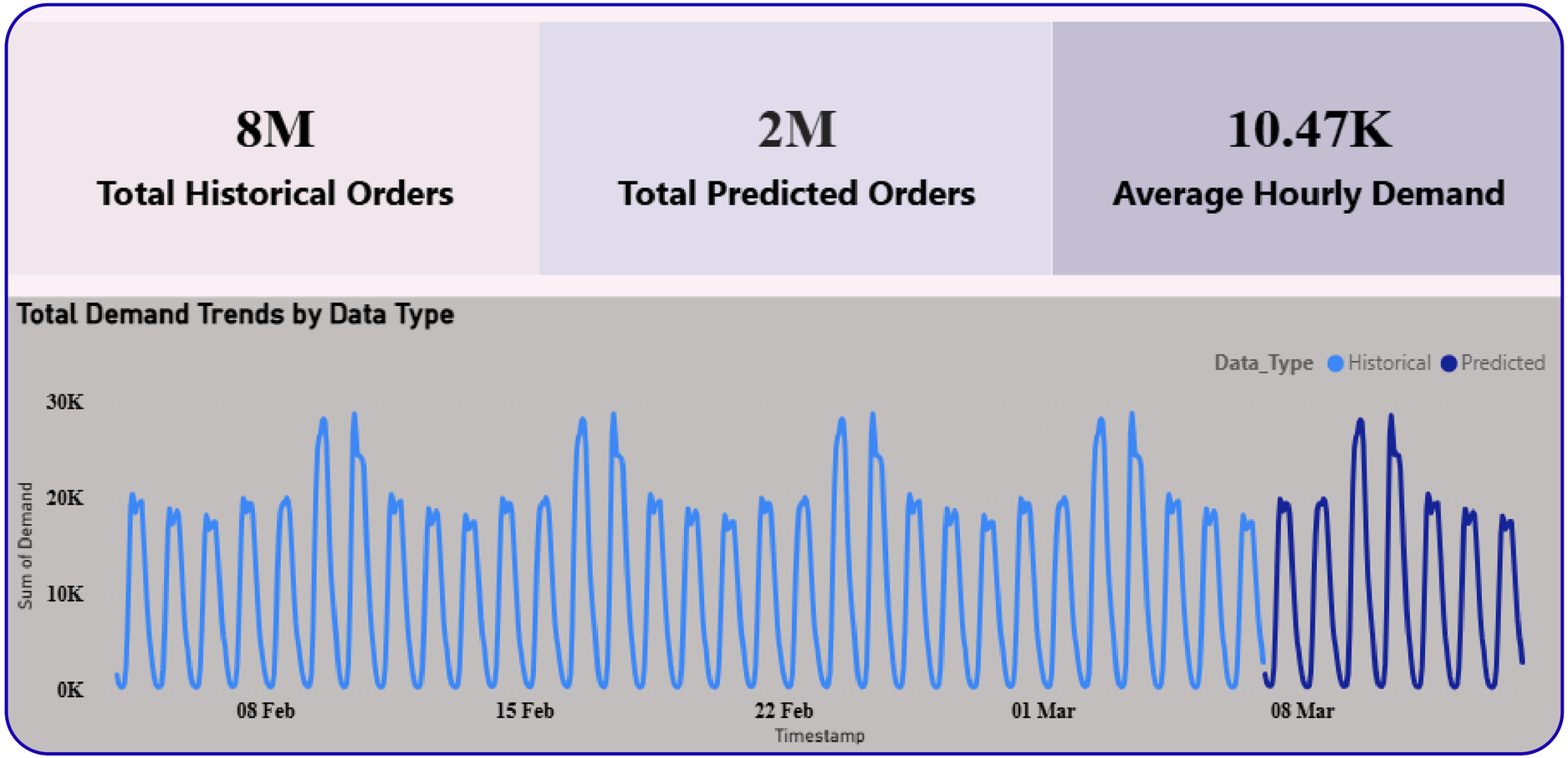
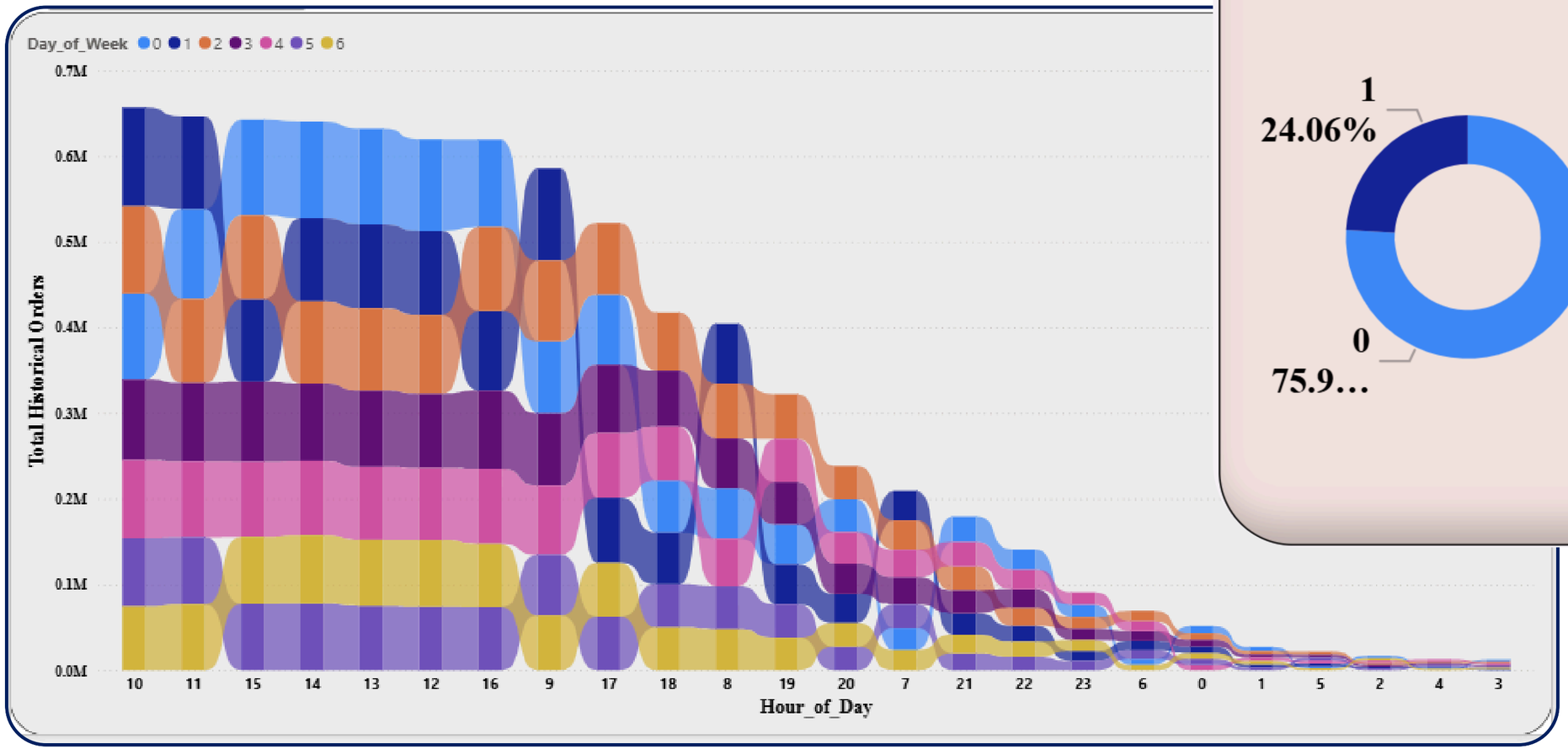


Image:1 Demand Trend Historical Vs Predicted

Result:



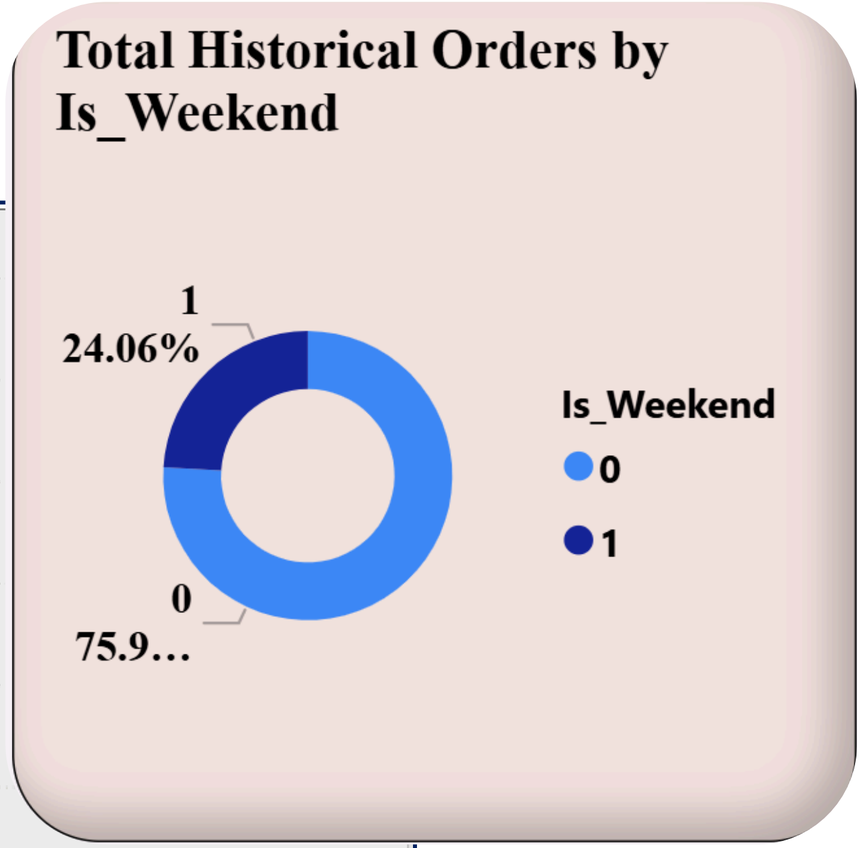


Image:2 Total Historical Orders BY Hour and Week

Conclusion:

- The system successfully transitions raw, chaotic order data into clear, actionable business intelligence.
- By combining the predictive power of Python (XGBoost) with the interactive visuals of Power BI, dark store managers can accurately prepare staff scheduling and inventory levels.
- The integration of Power BI's automated anomaly detection removes the need for manual data mining, saving time and immediately highlighting unusual business events.

Future scope:

- **Live API Integration:** Connecting the dashboard directly to a live inventory database for real-time, up-to-the-minute adjustments.

- **External Data Sources:** Incorporating hyper-local weather APIs (e.g., sudden rain forecasting) and local holiday calendars to improve the model's accuracy during unusual events.

- **Automated Reordering:** Scaling the system so that the AI not only predicts demand but automatically triggers purchase orders from central warehouses when predicted stock falls too low.

References:

- Instacart Market Basket Analysis Dataset (Kaggle).
- XGBoost Python Documentation for Time-Series Forecasting.
- Microsoft Power BI DAX and AI Visuals Documentation.

Project GitHub Link

Thank You

Q&A